Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 233 (2024) 892–901

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

Temporal Fusion of Time-Distributed VGG-16 and LSTM for Precise Action Recognition in Video Sequences

Athira K A[a], Divya Udayan J[b,*]

[a,b]Department of Computer Science and Engineering,Amrita School of Computing, Amrita Vishwa Vidyapeetham, Amritapuri,India
*Corresponding author email id: divyaudayanj@am.amrita.edu

Abstract

Autonomous systems play a vital role in creating an ambient environment, but for comprehensive human-robot interaction, the system must interpret and identify specific human behaviours. The demand for recognizing human activity from videos has significantly increased across diverse domains, such as medicine, intelligent video surveillance, human-robot interaction, and ambient-assisted environments. Deep learning has emerged as a pioneering technique for human activity recognition from videos due to its ability to learn features from input videos automatically. This research proposes a novel approach for recognizing activities in videos by leveraging the combined power of the VGG-16 convolutional neural network (CNN) and the Long Short-Term Memory (LSTM). To assess the efficiency of our suggested methodology, experiments are conducted on the widely used benchmark dataset UCF-50. The experimental results demonstrate that our VGG-16 and LSTM-based model outperforms state-of-the-art methods for activity recognition, achieving higher accuracy and robustness across various activity categories.

© 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: Activity Recognition,;Deep Learning;Visual Geometry Group-16(VGG-16) ;Long Short Term Memory(LSTM).

1. Introduction

Human activity recognition (HAR) is a rapidly evolving field in artificial intelligence and computer vision,[1] which aims to enable machines to comprehend and interpret human actions and behaviours from various sensory inputs, such as video, audio, and sensor data. To provide individualized and context-aware services in ambient-aided environments, the ability to recognize human behaviours from data streams is becoming increasingly important. The term ambient

1877-0509 © 2024 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application
10.1016/j.procs.2024.03.278

assisted living (AAL) describes incorporating technology into living areas to improve people's autonomy and quality of life, especially older people and those with disabilities. Ambient settings can be configured to react by utilizing intelligent systems that recognize human activity precisely. Significant progress has been made in deep learning approaches, specifically in Convolutional neural networks, also known as CNNs, and Recurrent neural networks, or (RNNs), which have fueled the development of algorithms for human activity recognition. [2]. These models have demonstrated exceptional performance in image and speech recognition tasks, which form the basis for activity recognition from visual and audio data. Identifying activities within a video sequence poses a formidable challenge due to variations in illumination conditions, object similarity across frames, and changes in the viewpoint for the same activity. Moreover, human activities can range from simple arm or leg actions to complex integrated movements involving the collective motion of both. For instance,[3] practicing yoga can be considered a complex integrated activity. While humans can easily comprehend actions in a video by grasping the contents and context of the frames, automating this process requires advanced techniques

Our methodology involves two key stages. Firstly, VGG-16 CNN is employed to extract high-level visual features from individual video frames. The pre-trained VGG-16 model has demonstrated its effectiveness in image recognition tasks, and here adapt it for video frame analysis to capture the spatial features relevant to the activity at hand. Secondly, feed the sequential feature representations obtained from the VGG-16 to an LSTM model, which excels at capturing temporal dependencies in data. By considering the temporal evolution of the visual features, the LSTM can learn the dynamic patterns inherent in video sequences, thereby enhancing the recognition accuracy for complex activities. The sequence of video frames is analysed in the proposed work, and spatial features are extracted from every third video frame using an already trained VGG-16 mode. Next, inorder to extract the sequential information from the video frame, the VGG-16 output is fed into an LSTM.

When compared to other models, our work aims to present a novel method using the Time Distributed VGG-16 in combination with an LSTM architecture. A thorough analysis of sequential data is made possible by this innovative model design, which permits the application of a layer to each temporal slice of an input sequence. A key element of this methodology is the Time Distributed Layer, which makes it easier to handle sequences of input data where the temporal dimension, which is indexed as the first dimension of the original input is important. It requires inputs with at least three dimensions to guarantee a strong handling of sequential data representation. Time-distributed VGG-16 is robust to variations in lighting, background, and camera motion. This means that it can be used to recognize activities in videos that are taken in a variety of condition.

2. Related Works

Various architectures exist in deep learning, but the Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) are the most prevalent models for vision-based human activity recognition. These models have demonstrated impressive performance on benchmark video datasets. Numerous recent studies have been conducted on activity recognition using hand-crafted features and neural networks. Among these approaches, deep learning has gained considerable popularity.

For example, researchers in [4] developed a 3D convolutional neural network (CNN) for action recognition, effectively capturing spatial and temporal information using 3D convolutions. In [5], an Adaptive Segmentation and Sequence Learning (ASSL) system was presented, which uses LSTM architecture to handle data from 3D joints in the human bod. LSTM layers were stacked to extract spatial and temporal information from frame sequences for action recognition. [6] used a different approach, employing two CNN architectures to process each frame, and 1x1 kernels were used to process intermediate layer outputs. They incorporated a 30-frame unrolled LSTM cell, joined with the CNN output during training. Another approach involved implementing 3D Convolutional Neural Networks with 3D motion cuboids for action detection [7] and conducting experiments on benchmark KTH and Weizmann datasets. While deep learning architectures excel in identifying video patterns due to their powerful feature extraction pipeline,

they also demand substantial training time and computational power. To address this, the authors of this paper proposed a method that processes every third frame, making it more efficient and less computationally complex. They combined a pre-trained time-distributed VGG-16 architecture with an LSTM model to recognize activities in videos efficiently. This paper [8] introduces a hybrid neural network architecture to classify human video activities. The architecture merges a layer of 3DCNN with a layer of ConvLSTM. The suggested fusion constitutes a potent deep-learning framework, albeit accompanied by various limitations that may influence its scalability, interpretability, data prerequisites, hyperparameter adjustment, and susceptibility to noise and incomplete data.

In [9], Alexnet was utilized for feature extraction, and the output was fed into a Deep Bidirectional LSTM. The model's efficiency was evaluated on different benchmark datasets. This paper [10] propose the use of a temporal convolutional neural network for the examination and categorization of human activities. The architecture introduced process brief videos, approximately 2 seconds in duration. This approach enables the simultaneous analysis of multiple frames within the video. Another approach introduced a two-stream CNN architecture [11], where the second stream concentrates on intensive optical flow while the first stream records spatial and temporal information between frames. To improve training, they combined two datasets. This [12] paper introduces an approach for fine-grained action retrieval in videos. The authors employ multiple parts-of-speech embeddings of spatial-temporal tokens, enabling the model to focus on different aspects of actions and achieve more precise recognition of fine-grained activities. In this research, [13] the authors propose a Temporal Pyramid Network that captures temporal dependencies at different scales. By aggregating features over multiple temporal levels, this model achieves robust recognition of long-range temporal patterns, thus enhancing the overall accuracy of action recognition.

[14] presented a novel architecture for video classification. The paper emphasizes the significance of transfer learning in boosting video understanding performance, particularly in scenarios where labeled data is limited. For the purpose of classifying face expressions in the ImageNet classification dataset [15], a lightweight vision transformer model called MobileVit—which outperforms existing lightweight CNN models is selected. [16] Introduced SlowFast Networks, a model designed to capture both spatial and temporal information efficiently. The dual-path architecture addresses challenges related to video recognition attaining cutting-edge outcomes and highlighting the significance of taking various speeds for various forms of information into consideration**.** [17] method involves consolidating local convolutional features throughout the video's entire spatial and temporal scope. By combining sophisticated two-stream networks with flexible spatiotemporal feature aggregation, this is accomplished. The resulting architecture is trainable end-to-end, enabling comprehensive video classification While slightly predating the specified timeframe, [18] work on Spatiotemporal Residual Networks remains influential. The paper introduces a novel architecture for video action recognition, emphasizing the significance of residual learning to effectively capture temporal and spatial correlations.

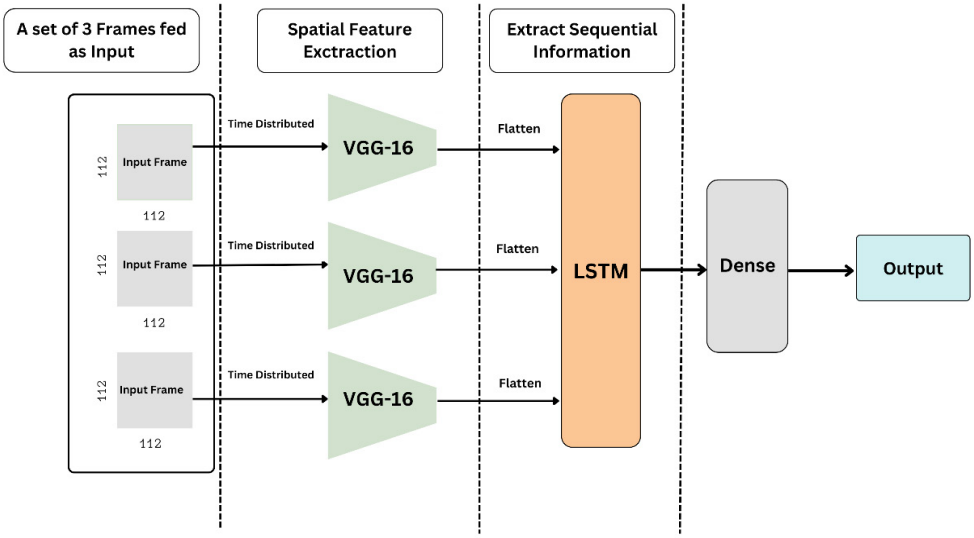


Fig. 1. Proposed framework for activity recognition using VGG-16 and LSTM

3. Proposed Framework

This part discuss the architecture suggested in detail, which includes activity recognition from the sequence of frames in video using time-distributed VGG-19 and LSTM. There are two parts to the complete action recognition process. Initially, take deep characteristics out of the frame using time-distributed VGG-16, where three frames are fed into the VGG-16 model at a time. The output from pre-trained VGG-16 architecture is given to the LSTM layers in the second part. The output layer of LSTM predicts the action performed in the vision data. The framework proposed is shown in Fig. 1. These two steps are discussed in a separate section.

3.1. Preparation and Feature Extraction

The CNNs are a powerful tool for representing and classifying images. When dealing with video data, each frame in the video is first represented by CNN features, and then sequential information between frames is collected using Long Short-Term Memory (LSTM). Videos may have varying frame rates, ranging from 40 to N frames per second, and within a given unit of time, there can be many repeating frames. The proposed framework processes every third frame for activity recognition to address the computational burden caused by processing redundant frames. It usually takes thousands of images to train a deep-learning model for images and a significant amount of processing power to alter the CNN model's weight. This framework uses the pre-trained VGG-16 architecture for feature extraction, which was trained on the ImageNet dataset with over 15 million images. The VGG-16 model was chosen due to its deep-layered architecture, which consists of 16 fully linked and convolutional layers and is well-known for its ability to capture complex visual representations. The pre-trained VGG-16 model was used to extract high-level features from the dataset using the principles of transfer learning. Through the network's propagation of input images, hierarchical and semantically significant features from the intermediary layers might be abstracted.

Instead of each input being connected to all neurons in the first layer, three frames are input into the network simultaneously. This helps avoid merging images since the pixel lists of all three images are sent to the first layer

together. Training separate neural networks for individual frames would be time-consuming as it would require multiple convolution flows, one for each input image. Moreover, these separate flows may result in different feature extraction, leading to inconsistency in detection. Also, if there are multiple images, the model should refrain from combining multiple images into one since the first layer will receive the complete pixel list of all the images. Three time-distributed VGG-16 networks are implemented to overcome this, and three frames of size 112x112x3 are fed into each network.

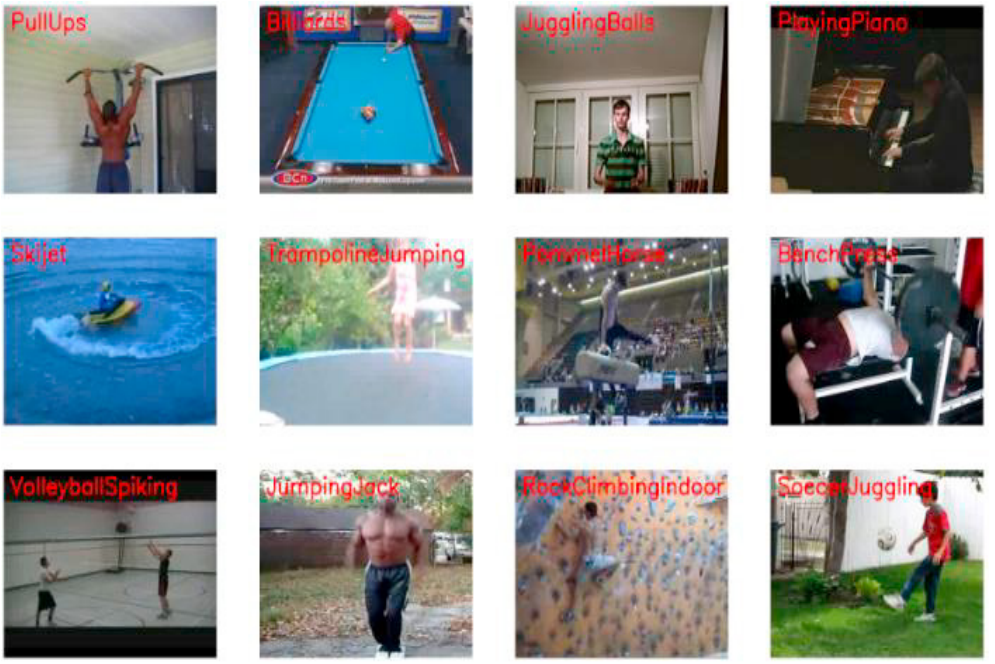


Fig. 2. Sample action categories of UCF-50 Dataset

3.2. Long Short-Term Memory (LSTM)

RNNs are designed to recognize underlying sequential patterns in spatial and temporal sequences. In the context of videos, which consist of sequential frames capturing movements in visual data, analyzing the sequence of frames is crucial for understanding the activity being performed. While RNNs can process such sequences, they face an issue known as the vanishing gradient problem when dealing with long-term data. This problem occurs when the RNN forgets earlier inputs in the sequence. To overcome the vanishing gradient problem and effectively learn long-term dependencies in sequences, LSTM is employed. LSTM's unique architecture, featuring input, output, and forget gates, allows for controlling and identifying long-term sequence patterns. By taking the output from the time-distributed VGG-16 layers as input, LSTM can accurately recognize the action performed by gathering sequential information from the input sequence. The Long Short-Term Memory (LSTM) network excels in handling sequential data, enabling the capture of contextual information crucial for understanding relationships between frames in a given activity context. Its expertise lies in discerning actions within sequences by leveraging temporal order and correlations among frames. Moreover, the LSTM's ability to incorporate historical data enhances its efficiency in identifying recurring patterns or transitions present in video data. Consequently, this strengthens the model's grasp on the sequential nature of activities, thereby improving its performance in recognizing actions. Combining LSTM with the VGG-16 model empowers the system not just to decipher spatial features from individual frames but also to understand the temporal connections among these frames. This cohesive fusion of spatial and temporal information significantly enhances the

accuracy and detail of activity recognition in video datasets, resulting in a more thorough and robust comprehension of intricate actions depicted in video sequences.

Action recognition within video sequences utilizes a fusion approach integrating the Time-Distributed VGG-16 and Long Short-Term Memory (LSTM) networks, constituting a sequential procedure encompassing feature extraction and temporal analysis. Initially, the VGG-16 architecture, recognized for its proficiency in image understanding, extracts spatial features autonomously from each frame in the video series. These features encapsulate crucial spatial details pertinent to action comprehension. Subsequently, the time-distributed methodology is employed, enabling the VGG-16 model to process frames iteratively, treating each frame as a sequential arrangement of spatial features. The LSTM network, acknowledged for its adeptness in capturing temporal relationships, receives these sequential spatial features and engages in temporal modeling. It scrutinizes inter-frame connections and assimilates extended temporal patterns within the video sequence. This fusion, amalgamating spatial information extraction via VGG-16 and LSTM's temporal modeling capabilities within a time-distributed framework, fosters a holistic grasp of spatial and temporal dynamics, facilitating precise action recognition within video sequences.

4. Experimental Results and Performance Evaluation

The dataset utilized in this experiment is UCF50, which serves as a benchmark dataset for action recognition. UCF50 comprises 50 distinct categories of videos featuring various realistic action scenarios. It is widely regarded as a highly renowned dataset for action recognition owing to its diverse and challenging nature. The UCF50 dataset covers a wide range of action categories, including 1) Interactions between humans and objects, 2) Actions that include solely body motion, 3) Interactions between humans, 4) Playing musical instruments, and 5) Engaging in sports activities. Fig. 2. Sample action categories of UCF-50 Dataset. Within these categories, numerous actions are captured in different contexts, such as sports activities, many of which occur against a similar background, often characterized by greenery. Additionally, several movies are recorded under different lighting conditions, body positions, and perspectives, which contributes to the complexity of the collection. One of the primary challenges of UCF50 is that the actions depicted are performed in real-life situations, making it distinct from other datasets where actors commonly act out actions. This aspect contributes to the dataset's realism and represents a unique characteristic for action recognition tasks.

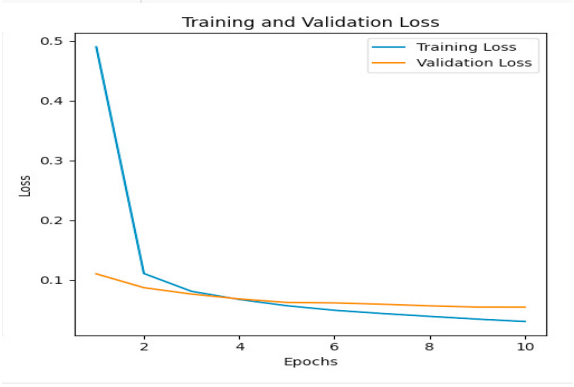


Fig. 3. Total loss versus total validation loss

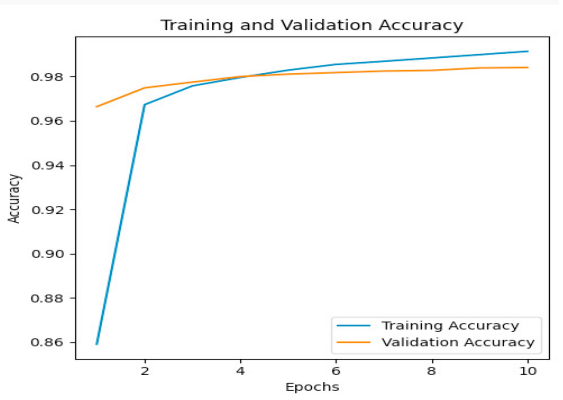


Fig. 4. Total accuracy versus total validation accuracy

A comprehensive experimental evaluation of the suggested framework is conducted with a well-known benchmark dataset for action recognition. The approach is carefully tested in this evaluation on a subset of 20% of the dataset's films, and it achieves an astonishingly high accuracy rate of 97%. The model is particularly trained on fifteen different classes that were carefully selected from the action detection dataset in order to carry out the experiment. The progression of the total loss vs the total validation loss is seen in Fig. 3. Throughout the training process, there is a constant downward trend in the loss measures for both training and validation. This decrease in losses shows that the model is getting better at learning and predicting things more accurately over time. It is clear that the model fits the data quite well when one looks at the increasing accuracy trends and the simultaneous decrease in losses seen in the training and validation sets. The model's ability to improve its prediction power over time and its consistent performance demonstrates its applicability and effectiveness in correctly identifying behaviours in the dataset. Fig. 4. plots the total accuracy against the total validation accuracy using the accuracy data acquired during the training and validation processes. First, the training accuracy starts out at a very high 85% and increases to a very high 99%. In a similar vein, the validation accuracy starts at a respectable 97% and rises to an astounding 98.4%. This increasing trajectory in accuracy demonstrates the model's strong learning process and its capacity to understand and categorise actions. Also, Fig 5. illustrates the confusion matrix, which provides insights into how well the model is performing across different classes. A method for visualising how well a machine learning model performs in classification tasks is the confusion matrix. The number of true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions made by the model are displayed for a number of classes. The instances in each column of the matrix correspond to a predicted class, and the examples in each row represent an actual class. The number of accurate predictions for each class is represented by the diagonal elements, which go from top-left to bottom-right, while misclassifications are represented by the off-diagonal elements. Fig.6. displays the model's projected actions based on the input video that was provided.

The effectiveness of the suggested method has been thoroughly assessed by direct comparison with two other frameworks: Long Term Recurrent Convolutional Neural Network (LRCN) and ConvLSTM, which blends the architectures of Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM). It is shown in Table 1. During this comprehensive comparison, a variety of criteria were used to evaluate the effectiveness of various frameworks, with accuracy acting as the main success metric. ConvLSTM and LRCN share the integration of recurrent and convolutional layers; hence, the combination of these processing modalities can pose challenges for the separation of temporal and spatial input. This could make it more difficult for the network to depict intricate temporal relationships or to capture fine-grained spatial characteristics. Better performance in certain action detection tasks is thus possible because to the Time Distributed VGG16 + LSTM architecture, which offers a more divided and effective way to manage temporal and spatial components independently.

The proposed method performs better than the LRCN and ConvLSTM framework and demonstrates improved accuracy in recognising and classifying actions or activities within the dataset. The comparative results, which have been laboriously assembled and presented in Table 1, offer a concise and understandable overview of how the proposed approach compares to the rival frameworks. Specifically, the table presents the accuracy metrics attained by each framework, emphasising the superior performance of the suggested approach in comparison to LRCN and ConvLSTM. This comprehensive comparison provides important information about the advantages of the proposed framework and places it as the best approach for accurate action recognition across the evaluated dataset.

Table 1. Comparison with other models

Method	#Epochs	Accuracy
ConvLSTM	10	74.25%
LRCN	10	92.86%
Proposed Approach (VGG-16 +LSTM)	10	98%

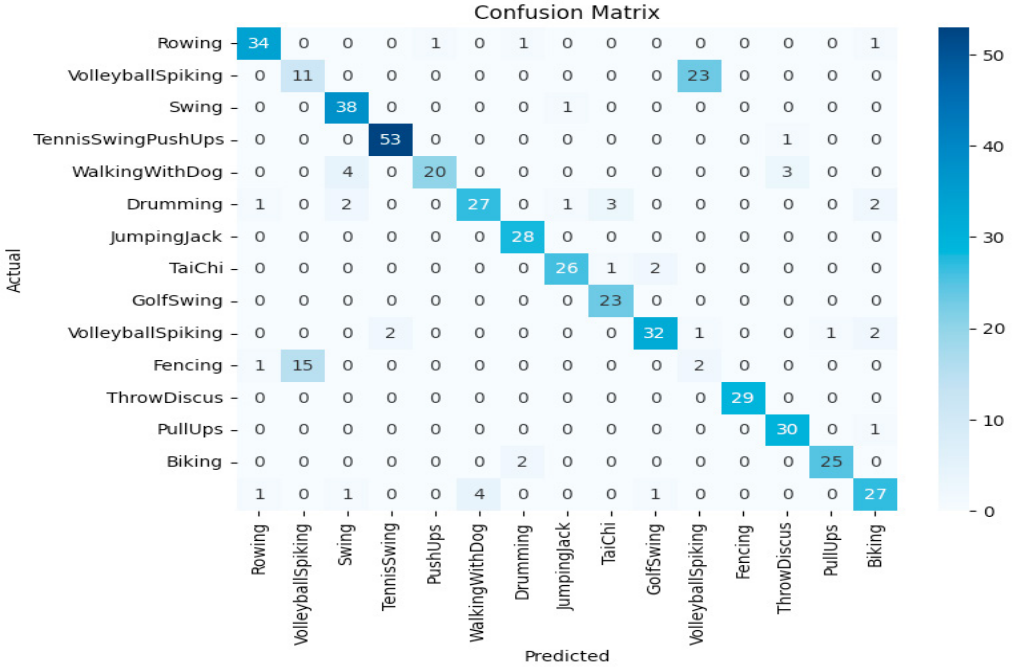


Fig 5. Confusion Matrix showing how well the model is performing on different classes



VGG-16 -LSTM Output
Action Predicted: Swing
Confidence: 0.36732780933380127
100% [00:19<00:00, 82.73it/s]



VGG-16 -LSTM Output
Action Predicted: Rowing
Confidence: 0.34824356735825415
100% [00:19<00:00, 82.73it/s]

Fig. 6. Shows the actions predicted by the model according to the given input video

5. Conclusion

This paper presents a novel approach for activity recognition by combining the deep learning models VGG-16 and LSTM. At first, features are extracted from every third frame using a pre-trained VGG-16 model that is capable of processing time-distributed data. These features are then fed into an LSTM to analyze complex sequential information between frames. By incorporating the time-distributed VGG-16 and LSTM, this work successfully decreased the computing burden of the model, making it feasible for real-time applications like video surveillance and human-computer interaction.

The empirical results highlight the strength and efficiency of our approach in accurately identifying activities in video datasets. The acquired accuracy and probability scores demonstrate that our proposed method surpasses other current state-of-the-art action recognition techniques, especially when assessed on the UCF50 dataset. The features of our technology make it an attractive solution for processing visual data and emphasize its potential as an essential component for smart systems. This approach considers the complete frame for feature extraction. In future endeavors, can enhance our methodology by exclusively targeting prominent areas within the frame for the purpose of extracting distinctive characteristics and recognizing activities. The focused investigation has the capacity to significantly improve the accuracy and effectiveness of our approach in managing intricate visual data.

References

[1] Soni, V., Yadav, H., Semwal, V. B., Roy, B., Choubey, D. K., & Mallick, D. K. (2023, January). A novel smartphone-based human activity recognition using deep learning in health care. In *Machine Learning, Image Processing, Network Security and Data Sciences: Select Proceedings of 3rd International Conference on MIND 2021* (pp. 493-503). Singapore: Springer Nature Singapore.

[2] Zhao, R., Ali, H., & Van der Smagt, P. (2017, September). Two-stream RNN/CNN for action recognition in 3D videos. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 4260-4267). IEEE.

[3] Ananth, G., & Anuradha, R. (2022, November). Yoga Posture Classification using Deep Learning. In *2022 International Conference on Futuristic Technologies (INCOFT)* (pp. 1-6). IEEE.

[4] Ji, Shuiwang & Xu, Wei & Yang, Ming. (2010). 3D Convolutional Neural Networks for Human Action Recognition. Pattern Analysis and Machine Intelligence, IEEE Transactions on. 35. 495-502. 10.1109/TPAMI.2012.59.

[5] Adama, D. A., Lotfi, A., & Ranson, R. (2021). Adaptive segmentation and sequence learning of human activities from skeleton data. *Expert Systems with Applications*, *164*, 113836.

[6]Yue-Hei Ng, J., Hausknecht, M., Vijayanarasimhan, S., Vinyals, O., Monga, R., & Toderici, G. (2015). Beyond short snippets: Deep networks for video classification. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4694-4702).

[7] Arunnehru, J., Chamundeeswari, G., & Bharathi, S. P. (2018). Human action recognition using 3D convolutional neural networks with 3D motion cuboids in surveillance videos. *Procedia computer science*, *133*, 471-477.

[8] Vrskova, R., Kamencay, P., Hudec, R., & Sykora, P. (2023). A New Deep-Learning Method for Human Activity Recognition. *Sensors*, *23*(5), 2816.

[9] Ullah, A., Ahmad, J., Muhammad, K., Sajjad, M., & Baik, S. W. (2017). Action recognition in video sequences using deep bi-directional LSTM with CNN features. *IEEE access*, *6*, 1155-1166.

[10] Andrade-Ambriz, Y. A., Ledesma, S., Ibarra-Manzano, M. A., Oros-Flores, M. I., & Almanza-Ojeda, D. L. (2022). Human activity recognition using temporal convolutional neural network architecture. *Expert Systems with Applications*, *191*, 116287.

[11] Simonyan, K., & Zisserman, A. (2014). Two-stream convolutional networks for action recognition in videos. *Advances in neural information processing systems*, *27*.

[12] Wray, M., Larlus, D., Csurka, G., & Damen, D. (2019). Fine-grained action retrieval through multiple parts-of-speech embeddings. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 450-459).

[13] Yang, C., Xu, Y., Shi, J., Dai, B., & Zhou, B. (2020). Temporal pyramid network for action recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 591-600).

[14] Diba, A., Fayyaz, M., Sharma, V., Karami, A. H., Arzani, M. M., Yousefzadeh, R., & Van Gool, L. (2017). Temporal 3d convnets: New architecture and transfer learning for video classification. *arXiv preprint arXiv:1711.08200*.

[15] Devika, R., & Divya Udayan, J. (2022). Lightweight Deep Learning Facial Expression Recognition Model. In *ICT Systems and Sustainability: Proceedings of ICT4SD 2022* (pp. 493-503). Singapore: Springer Nature Singapore.

[16] Feichtenhofer, C., Fan, H., Malik, J., & He, K. (2019). Slowfast networks for video recognition. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 6202-6211).

[17] Girdhar, R., Ramanan, D., Gupta, A., Sivic, J., & Russell, B. (2017). Actionvlad: Learning spatio-temporal aggregation for action classification. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 971-980).

[18] Christoph, R., & Pinz, F. A. (2016). Spatiotemporal residual networks for video action recognition. *Advances in neural information processing systems*, *2*.